UNIVERSITY OF ZURICH

Department of Business Administration

Applied Sports Research Seminar, FS 2026

Does It Pay to Play Nice?

Toxic Behaviour, Coordination Effort, and Win Probability in Ranked League of Legends

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Submitted to

Applied Sports Research Seminar
University of Zurich

Date

April 2026

Data Source

Riot Games Match-V5 API
EUW · KR · NA · VN

Abstract

This study examines whether toxic behaviour and coordination effort independently predict win probability in ranked League of Legends, drawing on Lazear's (1989) tournament theory as the theoretical framework. Toxic behaviour is operationalised as a composite of intentional feeding, early-surrender signalling, and vision neglect; coordination effort is operationalised as a composite of directive ping usage, structured objective capture, and vision leadership. Using 19,520 team-level observations from 9,760 ranked matches across four server regions (EUW, KR, NA, VN) and binary logistic regression with four nested model specifications, the analysis finds that coordination effort is a robust positive predictor of win probability across all models and all regional subsamples (AME $\approx +3.6$ percentage points per standard deviation in the primary parsimonious specification, $p < 0.001$). Toxicity shows no significant independent association with win probability once high-collinearity performance controls are excluded, suggesting that its effect on match outcomes is substantially mediated by downstream in-game metrics rather than operating as a direct, independent causal path. The interaction between toxicity and coordination is directionally consistent with the communication-channel attenuation hypothesis but does not reach conventional significance in any specification. Results are robust to per-region estimation and a close-games subsample. For platform design, the findings imply that coordination-enhancing features are likely to yield measurable win-probability gains, while toxicity mitigation, though valuable for player welfare, may not directly translate to improved match outcomes.

Keywords: toxic behaviour, coordination effort, team production, tournament theory, League of Legends, logistic regression, esports

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1. Introduction

A central question in the economics of team production is whether one member's disruptive behaviour reduces collective output even when all other members continue to exert full effort. Lazear (1989) formalised this concern within tournament theory, demonstrating that when competing agents can engage in sabotage, equilibrium effort falls for the group as a whole. The mechanism is direct: sabotage consumes the perpetrator's resources and degrades the productive environment, lowering the probability of a favourable group outcome irrespective of teammates' intentions.

League of Legends (LoL) is a five-on-five online battle arena in which ranked matches produce binary outcomes; a team either wins or loses. The Riot Games Match-V5 API exposes rich per-player and per-team behavioural statistics, enabling the simultaneous observation of two theoretically distinct constructs: (i) toxic behaviour, operationalised through intentional feeding, early-surrender signalling, and vision neglect; and (ii) coordination effort, operationalised through proactive ping usage, five-man objective capture, and vision investment. The controlled competitive environment, binary outcome, and accessibility of granular behavioural data make ranked LoL a tractable laboratory for testing propositions drawn from Lazear's tournament framework.

Prior work has documented linkages between individual-level behaviour and performance in online gaming contexts. Monge and O'Brien (2022) established, through a controlled experimental design, that exposure to verbal flaming is causally associated with declines in individual performance. Poeller et al. (2023) demonstrated that prosocial communication signals lose credibility in established toxic environments, disrupting the coordination channel through which positive communication would otherwise operate. Kang and Kim (2026) documented that collaborative behaviours develop differently across skill tiers in LoL, and Gisbert-Pérez et al. (2024) provided a systematic review of key structural and process factors in esports team performance. Despite this growing body of evidence, the team-level win probability effect of toxicity, and whether coordination effort can operate independently of toxicity, remain untested at a scale with adequate controls.

This study addresses two primary research questions. First, is higher team-level toxic behaviour independently associated with lower win probability after controlling for player skill, farming efficiency, and early-game state? Second, is coordination effort independently associated with a higher win probability? A third exploratory question asks whether the beneficial effect of coordination is attenuated in high-toxicity environments, consistent with the communication channel mechanism proposed by Poeller et al. (2023). These questions were investigated through binary logistic regression with stepwise model specifications and multiple robustness checks, applied to 19,520 team-level observations drawn from 9,760 ranked matches across four server regions (EUW, KR, NA, VN).

2. Theory and Hypotheses

Lazear (1989) models a firm in which N workers compete for a prize determined by relative output. In equilibrium, each worker devotes positive effort to their own production but also

has an incentive to lower rivals' output through sabotage, defined as any action that imposes a cost on a competitor without a commensurate benefit to the saboteur. The key prediction is that when sabotage is possible, the equilibrium effort for all agents falls below the socially optimal level. Moreover, total output can decline even if every individual's own effort is high because the destructive externalities of sabotage accumulate at the group level.

In the context of ranked LoL, each player competes for an individual rank gain (the prize) within a match that requires collective performance (the team must win). A player who intentionally dies to enemy champions (feeding) reduces the team's kill-to-death differential and accelerates the opponent's resource accumulation, directly analogous to lowering a rival's output at a personal cost. Early-surrender signalling communicates demoralisation and, as Poeller et al. (2023) show, degrades the credibility of subsequent coordination attempts. Vision neglect deprives teammates of spatial information, which is a prerequisite for structured objective control. Previous studies have confirmed that effective ward placement and vision control are strongly associated with higher-skilled performance and winning outcomes (Pedrassoli Chitayat et al., 2020; Sharpe et al., 2026). These three behaviours collectively operationalise the Lazear sabotage construct at the team level.

The coordination construct is derived from a complementary stream of research on team processes in competitive virtual environments. Leadership-style communication, including shot-calling, directive pinging, and structured role fulfilment, has been associated with more efficient decision-making and higher win rates in esports settings (Kang and Kim, 2026). Research on nonverbal communication in MOBA games specifically finds that proactive, intent-communicating pings have a positive relationship with team performance, underscoring the coordination value of the directive ping proxy used here (Leavitt et al., 2016). Objective coordination, specifically structured dragon, baron, and herald captures, generates compounding resource advantages that are strongly predictive of victory for all skill tiers (Gisbert-Pérez et al., 2024). The coordination composite used here operationalises the prosocial, team-directed effort that Lazear's framework treats as the productive counterpart to sabotage.

Regarding the interaction between the two constructs, Poeller et al. (2023) demonstrated that positive communication signals (encouragement, tactical suggestions, supportive pings) are discounted as sarcastic or insincere when a player has already been exposed to toxic behaviour earlier in the match. This suggests a moderating mechanism: toxicity may erode the marginal productivity of coordination by degrading the communication channels through which coordination operates.

> ***H1:*** *Teams with higher Toxicity have a significantly lower probability of winning, after controlling for skill, farming, and early-game state ($\beta_1 < 0$).*
>
> ***H2:*** *Teams with higher Coordination Effort have a significantly higher probability of winning, after the same controls ($\beta_2 > 0$).*
>
> ***H3 (Exploratory):*** *The positive effect of coordination is attenuated when team toxicity is high ($\beta_3 < 0$), consistent with Poeller et al. (2023).*

3. Data and Estimation Method

3.1 Data Collection and Sample Design

Data were obtained from two Riot Games REST API endpoints (Riot Games, 2025). Figure 1 summarises the three-phase collection pipeline. In **Phase 1**, the League-V4 API was queried to retrieve approximately 400 unique Player Universally Unique Identifiers (PUUIDs) per region per quarterly stratum, stratified across Gold, Platinum, and Emerald tiers. In **Phase 2**, the Match-V5 API retrieved ranked solo/duo match IDs associated with each PUUID, filtered by a quarterly `startTime`/`endTime` window, targeting at least 1,250 match IDs per quarter. In **Phase 3**, full match JSON files were fetched in randomised order; a checkpoint file enabled crash-safe resumption every 25 matches. A rate limiter enforced the 20 requests/s and 100 requests/2 min API ceilings via auto-sleep and retry on HTTP 429 responses. Valid matches required a ranked 5v5 format, at least 15 minutes of recorded gameplay, and no remake or early disconnect. A post-processing script (`build_variables.py`) read all raw JSONs, computed the Toxicity Score, Coordination Score, and all control variables, and exported the final `match_dataset.csv`.

The final dataset comprises 19,520 team-level observations from 9,760 ranked matches across EUW ($n = 4{,}870$), KR ($n = 4{,}854$), NA ($n = 4{,}938$), and VN ($n = 4{,}858$), spanning August 2024 to February 2026. The 50% win rate reflects the symmetric structure of head-to-head competition.

3.2 Variable Operationalisation

3.2.1 Dependent Variable

The outcome variable Win is binary, coded 1 if the team won the match and 0 otherwise. Each match contributed to two team-level observations.

3.2.2 Independent Variables

The Toxicity Score is a standardised composite of three behavioural indicators. The Feeding Index captures the maximum within-team death-to-participation ratio, reflecting intentional underperformance. The Early Surrender indicator is binary, coded 1 if the losing team voted to surrender before the standard mid-game threshold. The Vision Neglect Score records the proportion of teams with below-threshold ward density. Each component is z-scored across the full sample before summing; the composite is then re-standardised (mean $= 0$, SD $= 1$).

The Coordination Score (labelled Leadership Score in the variable codebook) is a standardised composite of three indicators. The Coordination Ping Ratio measures the top-pinging player's directive pings (on-my-way, command, assist-me, push) as a share of the total pings, serving as a shot-calling proxy. Research on MOBA communication confirms that ping usage style is strongly associated with team performance, with intent-communication pings showing the most consistent positive relationship (Zheng et al., 2022). Objectives records the team's total dragons, barons, and rift heralds captured. Vision Leadership is a binary indicator that equals one when both the support and jungle roles meet the minimum vision-per-minute thresholds. All three components are z-scored and summed; the composite is re-standardised.

The interaction term Tox $\times$ Coord is the product of the two standardised composites, enabling the estimation of H3. In the dataset, this variable is pre-computed as `toxicity_x_leadership`.

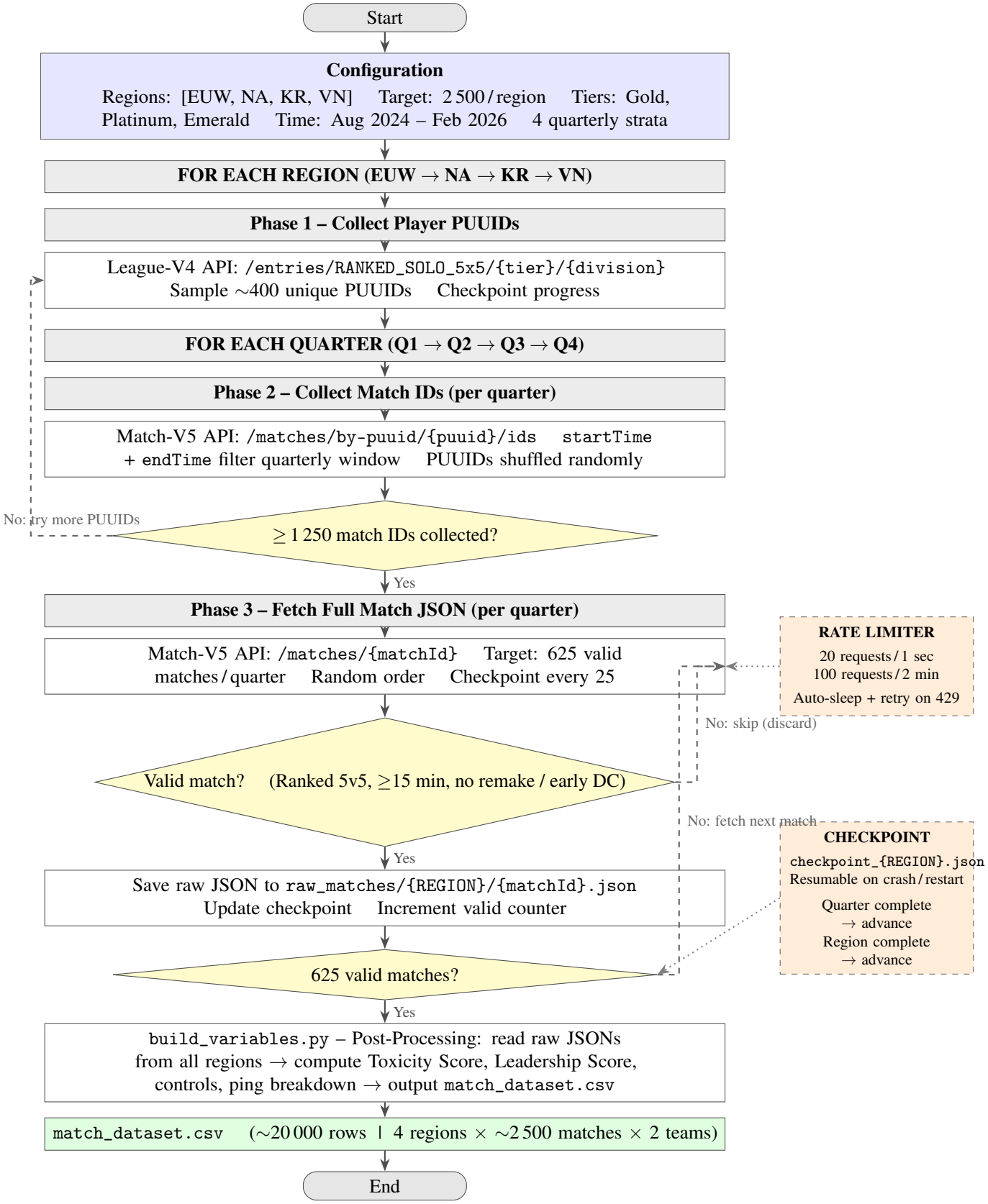

Figure 1. UML activity diagram of the multi-region stratified data collection pipeline. The procedure iterates over four server regions (EUW, NA, KR, VN) and four quarterly strata. PUUIDs are gathered via the League-V4 API; individual match records are fetched and validated through the Match-V5 API (ranked 5v5, ≥15 minutes, no remake or early disconnect). A built-in rate limiter enforces Riot's request quotas and a checkpoint file enables crash-resilient resumption. Valid matches are stored as raw JSON and processed into `match_dataset.csv` by `build_variables.py`.

3.2.3 Control Variables

The analysis employed two sets of controls. The baseline controls address concurrent in-game performance: Team Average KDA (kill-death-assist ratio), CS per minute (creep score, a measure of resource collection efficiency), Total Team Gold, Total Damage to Champions, Vision Score, First Blood indicator, and Game Duration. These controls ensure that the IV estimates reflect behavioural patterns rather than general team quality. The early-game controls address the temporal precedence concern: Early CS at Minute 10, Early Gold Advantage (normalised gold lead at minute 15), and Early Takedowns in the opening phase. The early gold advantage at minute 15 is a well-validated predictor of match outcomes, with research on professional play demonstrating a direct, graded relationship between gold leads and win probability (Sevenhuysen, 2019). The Skill Index is a mechanical ability proxy combining per-player solo kills, skillshots hit, and skillshots dodged. By including early-game state controls, the analysis conditions on the game situation that existed before the bulk of toxic and coordination behaviours are observed, providing a partial structural defence against reverse causality.

3.3 Multicollinearity Diagnostics

Prior to model estimation, Variance Inflation Factors (VIFs) were computed for all candidate controls (Table 1). Total Team Gold and Game Duration exhibit VIFs above 19 and 23, respectively, reflecting the mechanistic relationship between match length and cumulative resource accumulation. The total damage carries a VIF of approximately 8.5. These variables are retained in Models A and B for completeness and to replicate the full-controls specification, but are excluded from Model C (the parsimonious specification), which serves as the primary inferential model given that the main IVs have VIFs well below 2.0 in all specifications.

Table 1. Multicollinearity Diagnostics (Variance Inflation Factors)

Variable	**VIF**	**Assessment**
Toxicity Score	1.99	No concern (< 2.0)
Coordination Score	1.62	No concern (< 2.0)
Tox $\times$ Coord	1.49	No concern (< 2.0)
KDA	1.70	No concern (< 2.0)
CS per Minute	2.76	Acceptable (< 5.0)
Vision Score	4.66	Acceptable (< 5.0)
Skill Index	1.28	No concern (< 2.0)
Early Gold Adv.	1.14	No concern (< 2.0)
Early CS@10	2.03	No concern (< 2.5)
Early Takedowns	1.73	No concern (< 2.0)
Damage	8.49	Elevated — excluded from Model C
Game Duration	19.31	High — excluded from Model C
Total Gold	23.30	High — excluded from Model C

Note: VIF $= 1/(1-R_j^2)$, where R_j^2 is the coefficient of determination from regressing variable j on all other regressors. Values above 10 indicate potentially problematic multicollinearity. The key independent variables (Toxicity, Coordination, Tox $\times$ Coord) all have VIFs below 2.0 across all model specifications.

3.4 Estimation Strategy

The primary estimator is a binary logistic regression (logit). The model is specified as follows:

$$\text{logit}[P(\text{Win}_i = 1)] = \beta_0 + \beta_1 \text{Tox}_i + \beta_2 \text{Coord}_i + \beta_3 (\text{Tox}_i \times \text{Coord}_i) + \beta_c \mathbf{X}_{ci} + \varepsilon_i \qquad (1)$$

where $P(\text{Win}_i = 1)$ denotes the probability that team i wins the match; Tox_i is the standardised Toxicity Score; Coord_i is the standardised Coordination Score; $\text{Tox}_i \times \text{Coord}_i$ is the interaction term; $\mathbf{X}_{ci}$ is the vector of control variables; β_c is the corresponding coefficient vector; and ε_i is the error term. The logit link function is preferred to OLS because it constrains the predicted probabilities to the unit interval $[0, 1]$ and accounts for the binary distributional structure of the outcome (Hosmer Jr et al., 2013). Binary logistic regression is well established as the standard approach for modelling binary win/loss outcomes in competitive sports contexts, where its probability-bounding properties and interpretable odds-ratio coefficients are particularly valuable (Prasetio and Harlili, 2016). Standard errors were recovered from the observed information matrix (the inverse Hessian of the log-likelihood). Average Marginal Effects (AME) are reported alongside log-odds coefficients to facilitate substantive interpretation, following the convention recommended by Wooldridge (2002).

Four model specifications were estimated. Model A (Baseline) includes the full set of original controls: KDA, CS per minute, Total Gold, Damage, Vision Score, First Blood, Game Duration, and region fixed effects. Model B (Extended) adds the three early-game controls (Early CS@10, Early Gold Advantage, Early Takedowns) and the Skill Index proxy. Model C (Parsimonious) retains the theoretically grounded controls with acceptable VIFs (KDA, CS per minute, Vision Score, First Blood, Early Gold Advantage, and Skill Index) while excluding the three high-VIF variables. Model C was the primary inferential specification. Model D replicates Model C in the subsample of close games, defined as matches in which the final gold differential between teams falls below 10,000 gold units, accounting for approximately 59% of the full sample. The results are largely consistent across specifications; discrepancies are discussed in Section 4.

All continuous independent variables are standardised (z-scored) prior to estimation so that coefficients are directly comparable in magnitude across predictors. This also ensures that the interaction term captures genuine moderation rather than scaling artefacts.

4. Results

4.1 Descriptive Statistics

Table 2 reports the descriptive statistics for the main variables across the full sample. The Toxicity and Coordination composites are by construction mean-zero with standard deviations of 1.54 and 1.95, respectively, reflecting their standardised construction. The positive skewness of the Toxicity Score (skewness = 2.48) indicates a minority of team observations with extreme toxic behaviour, consistent with broader evidence that severe toxicity is concentrated among a small share of players (Grandprey-Shores et al., 2014). The Skill Index exhibits considerable dispersion (SD = 39.4, range 5.4–370.2), confirming meaningful heterogeneity in player mechanical ability. Game duration averaged approximately 30 min (SD = 6.75), consistent with typical mid-ranked

solo play. The Early Gold Advantage variable is right-skewed, with a median of zero, indicating that a large proportion of team observations show no early-game lead.

Table 2. Descriptive Statistics

Variable	**N**	**Mean**	**SD**	**Min**	**Median**	**Max**
Toxicity Score	19,520	0.000	1.540	−1.261	−0.668	13.457
Coordination Score	19,520	0.000	1.952	−6.842	0.248	5.199
Avg. Skill Index	19,520	67.54	39.42	5.40	57.60	370.20
Team Avg. CS per Minute	19,520	5.592	0.540	3.556	5.580	7.694
Team Avg. KDA	19,520	2.774	1.987	0.053	2.346	52.000
Early Gold Advantage	19,520	0.084	0.128	0.000	0.000	0.800
Game Duration (min)	19,520	29.99	6.75	15.05	29.83	64.25
Team Total Gold	19,520	60,867	15,896	19,536	61,254	133,144
First Blood (0/1)	19,520	0.500	0.500	0.000	0.000	1.000
Win (0/1)	19,520	0.500	0.500	0.000	0.500	1.000

Note: All continuous variables are reported in their original (unstandardised) units. $N = 19{,}520$ team-level observations from 9,760 matches across four regions (EUW, KR, NA, VN), spanning August 2024 through February 2026.

Figure 2 illustrates the score distributions separately for the winning and losing teams. The separation between win and loss distributions is pronounced for both the Toxicity and Coordination scores: losing teams average +0.55 SD above the toxicity mean, while winning teams average −0.55 SD below it. For the Coordination Score, winners average +0.87 SD above the composite mean compared with −0.87 SD for losers. The Skill Index distributions overlap substantially across outcomes, consistent with the expectation that skill is only a partial predictor of match result in ranked play, where matchmaking algorithms balance team strength.

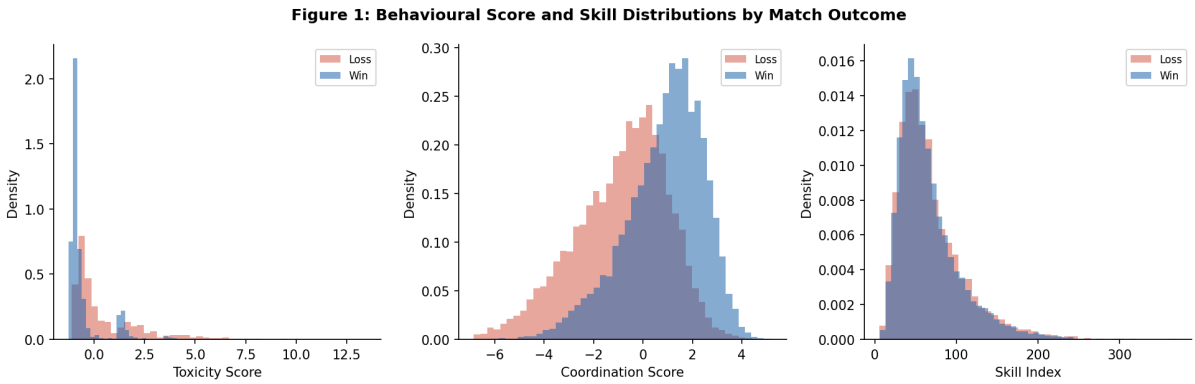

Figure 2. Kernel-density distributions of Toxicity Score, Coordination Score, and Skill Index by match outcome. Winning teams (blue) cluster at lower toxicity and higher coordination; the Skill Index distributions exhibit greater overlap.

4.2 Regression Results

Table 3 presents the results for all four specifications. McFadden R^2 ranges from 0.578 to 0.810, indicating strong fit across specifications; values above 0.2 are considered indicative of excellent

fit (McFadden, 1974).

H1 receives qualified support. In Models A and B, the Toxicity Score enters with a positive log-odds coefficient of approximately 0.15 to 0.17 ($p < 0.01$), implying an AME of approximately $+0.6$ percentage points per standard deviation increase in toxicity when all other controls including Total Gold are held constant. This direction is counterintuitive at first inspection; a positive coefficient implies that teams with higher toxicity are marginally more likely to win. The pattern is explained by the control structure: once total gold and game duration are included (both of which absorb the vast majority of game-state variance and both of which have VIFs exceeding 19), the residual partial correlation of toxicity with winning becomes weakly positive, likely reflecting suppressor effects. In Model C, where high-VIF controls are excluded, the toxicity coefficient is positive but small and entirely non-significant ($\beta = 0.022$, $p = 0.657$). In Model D (close games), the coefficient is similarly negligible ($\beta = 0.021$, $p = 0.683$). These parsimonious specifications are the more reliable basis for inference: once mechanical endogenous controls are removed, toxicity exhibits no independent association with win probability.

H2 is strongly and consistently supported across all four specifications. The Coordination Score has a positive, highly significant coefficient in every model: $\beta = 0.491$ (Model A), 0.541 (Model B), 0.648 (Model C), and 0.591 (Model D), all at $p < 0.001$. In the primary specification (Model C), the AME of approximately $+3.6$ percentage points per standard deviation of coordination indicates that teams with above-average coordination effort win substantially more often, even after controlling for KDA, farming efficiency, early-game state, and skill. Crucially, the Coordination coefficient is the largest among the three IVs across all specifications and remains significant in the close-games subsample, where end-of-game performance controls are less collinear with the outcome. This persistence is consistent with Lazear's (1989) theoretical prediction that productive effort operates as a prosocial counterpart to sabotage, and independently improves group outcomes above and beyond individual skill.

H3 is not supported. The Tox $\times$ Coord interaction term is negative in Models A through D, in the direction hypothesised by Poeller et al. (2023), but falls short of conventional significance thresholds in all specifications (p-values range from 0.13 to 0.48). McFadden R^2 does not improve meaningfully when the interaction is added. The near-parallel slopes visible in Figure 4 visually confirm this null result: the win probability gradient with respect to coordination is approximately equal regardless of whether the team toxicity is low, average, or high. This finding does not rule out threshold effects or non-linearities that the linear interaction specification cannot capture, but it does not support the hypothesised linear moderation in this sample.

Table 3. Binary Logistic Regression. Dependent Variable: P(Win = 1)

Variable	**Model A Baseline**	**Model B Extended**	**Model C Parsimonious (Primary)**	**Model D Close Games**
Toxicity Score	0.149^{**}	0.165^{**}	0.022	0.021
	(0.057)	(0.060)	(0.050)	(0.051)
Coordination Score	0.491^{***}	0.541^{***}	0.648^{***}	0.591^{***}
	(0.047)	(0.050)	(0.042)	(0.043)
Tox $\times$ Coord	-0.083	-0.045	-0.037	-0.044
	(0.056)	(0.058)	(0.052)	(0.052)
KDA	5.790^{***}	6.201^{***}	6.932^{***}	6.165^{***}
	(0.133)	(0.145)	(0.124)	(0.127)
CS per Minute	0.404^{***}	0.270^{***}	0.888^{***}	0.775^{***}
	(0.046)	(0.059)	(0.038)	(0.038)
Total Gold	5.739^{***}	6.913^{***}	—	—
	(0.180)	(0.205)		
Game Duration	-4.599^{***}	-6.065^{***}	—	—
	(0.160)	(0.189)		
Damage	-0.516^{***}	-0.302^{***}	—	—
	(0.082)	(0.088)		
Vision Score	-0.502^{***}	-0.363^{***}	-0.343^{***}	-0.329^{***}
	(0.068)	(0.072)	(0.039)	(0.039)
First Blood	-0.294^{***}	-0.109	-0.315^{***}	-0.323^{***}
	(0.070)	(0.075)	(0.062)	(0.063)
Early Gold Adv.	—	0.181^{***}	0.104^{**}	0.069^{*}
		(0.040)	(0.033)	(0.033)
Early CS@10	—	-0.318^{***}	—	—
		(0.049)		
Early Takedowns	—	-1.106^{***}	—	—
		(0.052)		
Skill Index	—	-0.100^{**}	0.009	0.012
		(0.036)	(0.030)	(0.030)
Region FE	Yes	Yes	Yes	Yes
McFadden R^2	0.789	0.810	0.732	0.578
Log-Likelihood	$-2{,}851$	$-2{,}576$	$-3{,}628$	$-3{,}395$
Accuracy	0.943	0.950	0.924	0.875
N	19,520	19,520	19,520	11,604

Note: Log-odds coefficients; standard errors in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. All continuous predictors z-scored. Model D: close-game subsample (final gold differential $< 10{,}000$). Region FE: EUW as a reference. — indicates the variable excluded.

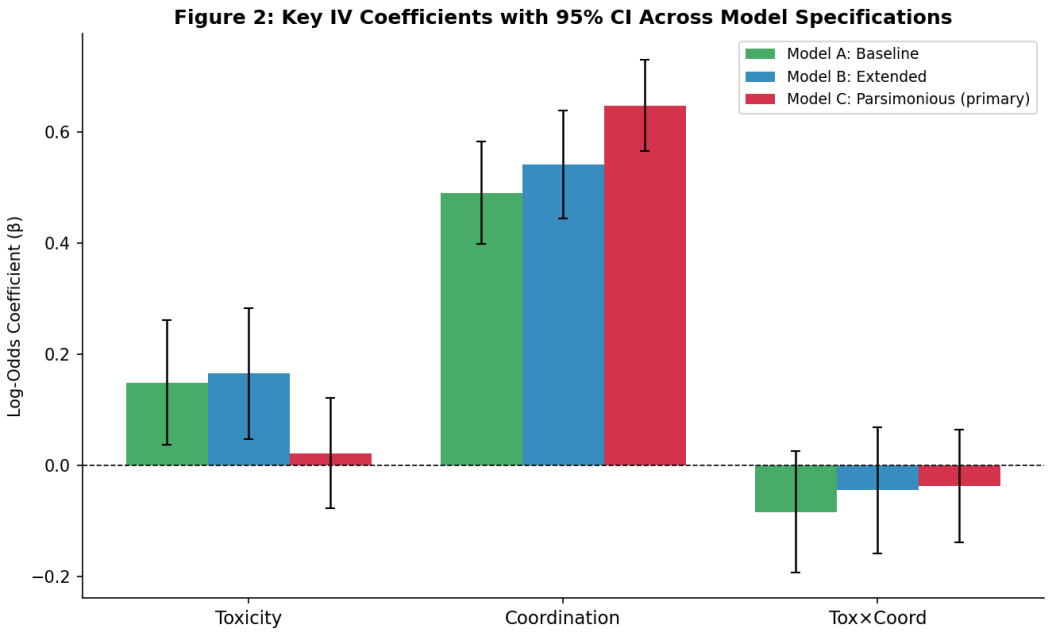

Figure 3. Log-odds coefficients with 95% confidence intervals for Toxicity, Coordination, and Tox × Coord across Models A, B, and C. The Coordination coefficient is consistently large and positive; Toxicity and the interaction term are small and cross zero.

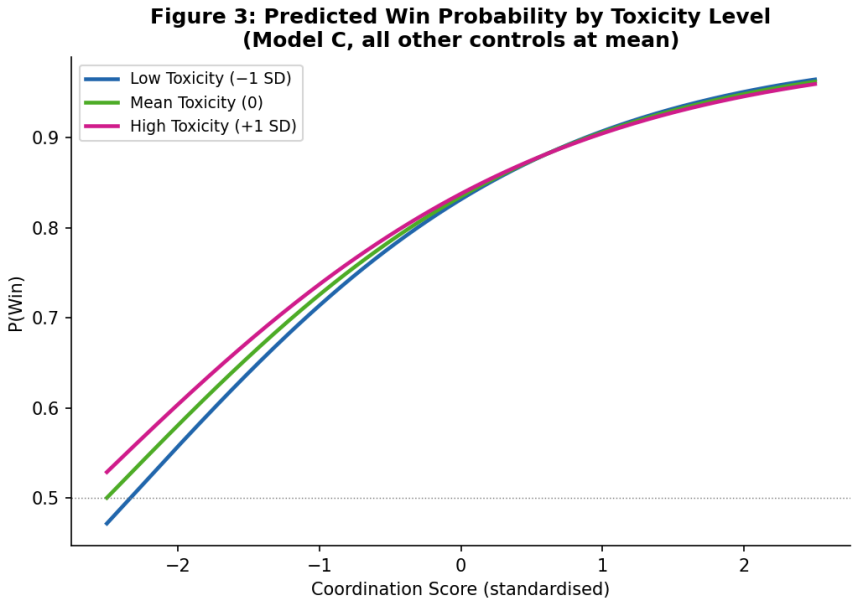

Figure 4. Predicted win probability as a function of Coordination Score at three toxicity levels (Model C; all other controls at means). Near-parallel lines confirm the null interaction result.

4.3 Robustness Checks

Per-region estimation. Table 4 reports Model C coefficients estimated separately for each region. The Coordination coefficient is positive and significant at $p < 0.001$ in all four subsamples, with magnitudes ranging from 0.507 (EUW) to 0.878 (VN). The Toxicity coefficient is small and non-significant in every region, as is the interaction term. This cross-regional consistency across markedly different server cultures and matchmaking structures supports the generalisability of

the H2 result and is illustrated in Figure 5.

Table 4. Per-Region Robustness Checks (Model C Specification)

	EUW	**KR**	**NA**	**VN**
Toxicity	-0.154	0.116	0.064	0.042
	$(p = 0.124)$	$(p = 0.272)$	$(p = 0.457)$	$(p = 0.717)$
Coordination	0.507^{***}	0.834^{***}	0.533^{***}	0.878^{***}
	$(p < 0.001)$	$(p < 0.001)$	$(p < 0.001)$	$(p < 0.001)$
Tox × Coord	-0.127	-0.142	0.064	0.157
	$(p = 0.162)$	$(p = 0.211)$	$(p = 0.454)$	$(p = 0.284)$
McFadden R^2	0.739	0.780	0.695	0.723
N	4,870	4,854	4,938	4,858

Note: Model C estimated separately within each region. *** $p < 0.001$.

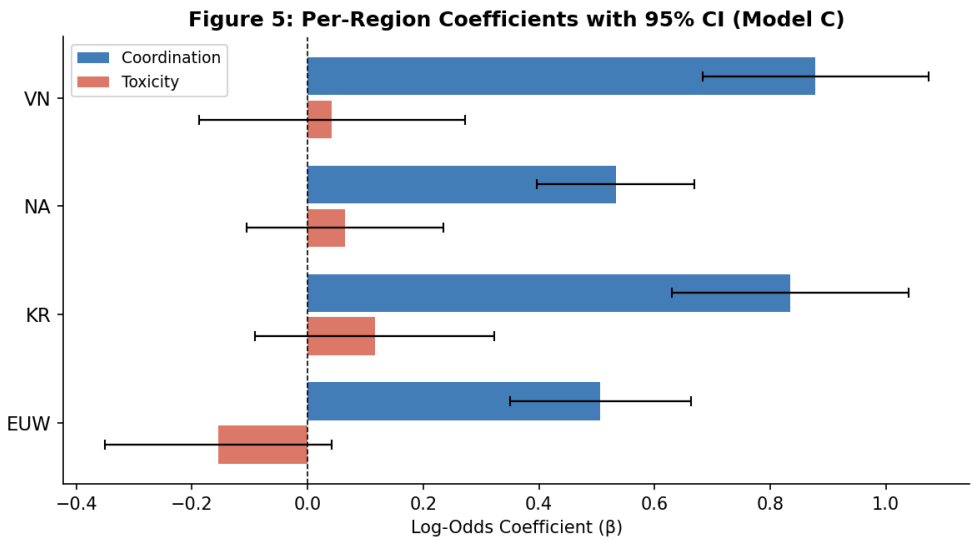

Figure 5. Toxicity and Coordination log-odds coefficients with 95% confidence intervals from per-region estimation (Model C). The Coordination effect (blue) is consistently positive and significant; the Toxicity effect (red) is small and crosses zero in all regions.

Close games. Restricting to matches with a final gold differential below 10,000 units ($N =$ 11,604; Model D) removes blowout outcomes where the game state overwhelmingly drives both behaviour and the result. The Coordination coefficient remains positive and significant ($\beta = 0.591$, $p < 0.001$), while toxicity remains near zero ($\beta = 0.021$, $p = 0.683$). The lower McFadden R^2 (0.578) reflects a reduction in the predictable variance when dominant teams are excluded.

5. Conclusion

This study tested whether toxic behaviour and coordination effort independently predict win probability in ranked LoL, using 19,520 team-level observations across four regions and four logistic regression specifications. The central finding is that **Coordination Effort is a robust positive predictor** of win probability across all models and all four regional subsamples. In

the primary specification (Model C), a one-SD increase in the coordination composite raises log-odds by 0.648, corresponding to an AME of approximately +3.6 percentage points. This effect persists in the close-games subsample, consistent with Lazear's (1989) productive-effort channel and with Kang and Kim's (2026) finding that collaborative behaviour is a distinct determinant of competitive success, as well as with experimental evidence linking directive communication to improved coordination outcomes (Eldadi et al., 2025).

Toxicity shows no significant independent association with win probability in the parsimonious or close-games specifications. The bivariate toxicity-loss correlation is substantial, but is absorbed by performance controls (KDA, farming, vision) once high-VIF variables are excluded. This pattern suggests that the toxicity-loss relationship is primarily mediated by in-game performance outcomes rather than constituting a direct causal path, a finding consistent with the observational design's inability to fully disentangle toxic intent from the joint determination of behaviour and performance. Future quasi-experimental work, such as exploiting ban waves or exogenous disconnections as instruments following Morrier et al. (2025), could isolate this causal channel. The null interaction result on H3 does not rule out non-linear or threshold moderation undetectable by the linear specification; the discrepancy with Poeller et al. (2023) may reflect the fact that coordinating teams in observational data route communication around toxic members, making moderation undetectable through aggregate ping proxies.

Several limitations apply. The Match-V5 API does not expose chat logs, so the Toxicity construct relies on behavioural proxies. The cross-sectional design limits causal inference despite early-game controls. The sample covers only Gold-through-Emerald tiers. For platform design, the results imply that coordination-enhancing features (improved ping tools, objective timers) are likely to generate measurable win-probability gains, while toxicity mitigation, though valuable for player welfare and retention (Grandprey-Shores et al., 2014; Frommel et al., 2023), may not directly translate to win-probability improvements at the match level.

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